

Verification of Probing Security for Masked Circuits

A Constructive Approach and Theoretical Study

Kağan Gülsüm

July 21, 2026

Sabancı University — M.Sc. Thesis Defense

Supervisor: *Asst. Prof. Süha Orhun Mutluergil*

The story of this talk

1. **Problem:** real devices leak; measuring power or timing can recover secret keys. **Standard fix:** masking; compute on randomized shares, never on secrets.

The story of this talk

1. **Problem:** real devices leak; measuring power or timing can recover secret keys. **Standard fix:** masking; compute on randomized shares, never on secrets.
2. **The question:** was the masking done correctly? “Correct” means no d observed wires can tell two secrets apart.

The story of this talk

1. **Problem:** real devices leak; measuring power or timing can recover secret keys. **Standard fix:** masking; compute on randomized shares, never on secrets.
2. **The question:** was the masking done correctly? “Correct” means no d observed wires can tell two secrets apart.
3. **It is provably hard:** as hard as counting solutions over all input combinations, so no tool can simply compute the answer.

The story of this talk

1. **Problem:** real devices leak; measuring power or timing can recover secret keys. **Standard fix:** masking; compute on randomized shares, never on secrets.
2. **The question:** was the masking done correctly? “Correct” means no d observed wires can tell two secrets apart.
3. **It is provably hard:** as hard as counting solutions over all input combinations, so no tool can simply compute the answer.
4. **Our route:** prove security by re-arranging the randomness until the secrets visibly cancel. Coppola does this in Lean 4, with a rewrite engine, a combinatorial search, and reusable certificates.

The story of this talk

1. **Problem:** real devices leak; measuring power or timing can recover secret keys. **Standard fix:** masking; compute on randomized shares, never on secrets.
2. **The question:** was the masking done correctly? “Correct” means no d observed wires can tell two secrets apart.
3. **It is provably hard:** as hard as counting solutions over all input combinations, so no tool can simply compute the answer.
4. **Our route:** prove security by re-arranging the randomness until the secrets visibly cancel. Coppola does this in Lean 4, with a rewrite engine, a combinatorial search, and reusable certificates.
5. **Result:** soundness proofs, gadgets verified that a comparable tool false-flags, and competitive performance.

Motivation & Model

A cipher can be secure on paper and insecure on the bench.

- A real device leaks through power draw, timing, electromagnetic emanation; all of which vary with the *data* it handles.
- Differential power analysis (Kocher et al. 1999) correlates many power traces against a guess of an intermediate value, and may recover the key.

So, the insecurity lies not in the mathematics behind the cipher, but in the way the hardware computes it.

This is not a theoretical worry. DPA has recovered keys from deployed smartcards, and timing side channels underlie the Spectre and Meltdown attacks against modern processors.

RESEARCH

Spectre haunts CPUs again: VMSCAPE vulnerability leaks cloud secrets

AMD Zen hardware and Intel Coffee Lake affected

 Thomas Claburn

4 

Published Thu 11 Sept 2025 // 15:00 UTC



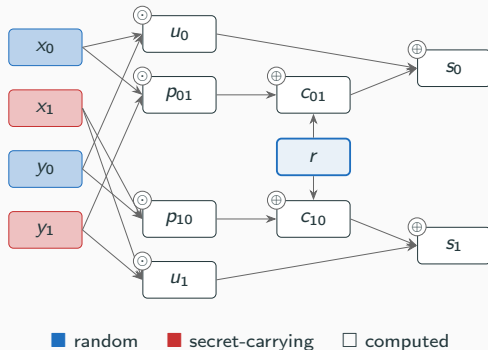
We study **arithmetic circuits** over the two-element field $\mathbb{F}_2$, built from two-input $\oplus$ (XOR) and $\odot$ (AND) gates.

Inputs are partitioned into three classes

- **secrets** $\mathcal{S}$ — the key material we protect,
- **randoms** $\mathcal{R}$ — fresh uniform bits (the masks), drawn together as a *tape* $\rho \in \{0, 1\}^{\mathcal{R}}$,
- **publics** $\mathcal{P}$ — known to everyone, held fixed.

Masking, and our running example

Boolean masking (Chari et al. 1999): split each secret into shares, $x = x_0 \oplus x_1$ with x_0 uniformly random, and compute only on shares, never on secrets.



DOM-AND (Groß et al. 2016): a masked AND gate, two shares per input, one fresh mask r .

- outputs re-share $s_0 \oplus s_1 = x \odot y$,
- cross terms p_{01}, p_{10} mix shares of *both* secrets,
- r re-masks them before the outputs.

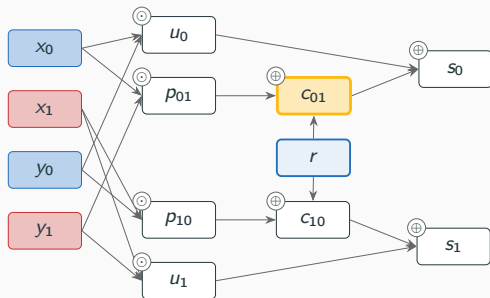
The probing model

Reasoning about noisy traces directly is unwieldy. Luckily we have an abstraction (Ishai et al. 2003):

***d*-probing security**

An adversary picks any d wires (probe; observation set) $O = \{w_1, \dots, w_d\}$ and is handed their *exact* values. The circuit is d -probing secure when, for every such choice and every two secret assignments a, b , the observed values are identically distributed over the tape ρ :

$$\mathcal{L}_a(w_1, \dots, w_d) = \mathcal{L}_b(w_1, \dots, w_d).$$



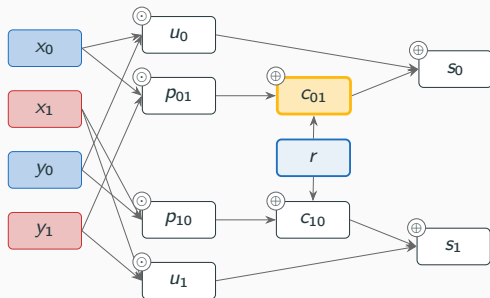
The probing model

Reasoning about noisy traces directly is unwieldy. Luckily we have an abstraction (Ishai et al. 2003):

d -probing security

An adversary picks any d wires (probe; observation set) $O = \{w_1, \dots, w_d\}$ and is handed their *exact* values. The circuit is d -probing secure when, for every such choice and every two secret assignments a, b , the observed values are identically distributed over the tape ρ :

$$\mathcal{L}_a(w_1, \dots, w_d) = \mathcal{L}_b(w_1, \dots, w_d).$$



- Security becomes a *combinatorial* property, decidable in principle.

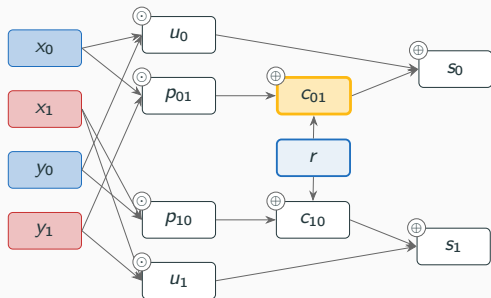
The probing model

Reasoning about noisy traces directly is unwieldy. Luckily we have an abstraction (Ishai et al. 2003):

***d*-probing security**

An adversary picks any d wires (probe; observation set) $O = \{w_1, \dots, w_d\}$ and is handed their *exact* values. The circuit is d -probing secure when, for every such choice and every two secret assignments a, b , the observed values are identically distributed over the tape ρ :

$$\mathcal{L}_a(w_1, \dots, w_d) = \mathcal{L}_b(w_1, \dots, w_d).$$



- Security becomes a *combinatorial* property, decidable in principle.
- The abstraction is faithful: probing security implies noisy-leakage security (Duc et al. 2014).

The problem

Exact verification of d -probing security

Decide, for every one of the $\sum_{i \leq d} \binom{|\mathcal{W}|}{i}$ observation sets O , whether it is *independent of the secrets*:

$$\mathcal{L}_a(O) = \mathcal{L}_b(O) \quad \text{for all } a, b \in \{0, 1\}^{\mathcal{S}}.$$

- Secure $\Rightarrow$ every observation set accounted for,
- Insecure $\Rightarrow$ a specific leaking observation as witness.

The problem

Exact verification of d -probing security

Decide, for every one of the $\sum_{i \leq d} \binom{|\mathcal{W}|}{i}$ observation sets O , whether it is *independent of the secrets*:

$$\mathcal{L}_a(O) = \mathcal{L}_b(O) \quad \text{for all } a, b \in \{0, 1\}^{\mathcal{S}}.$$

- Secure $\Rightarrow$ every observation set accounted for,
- Insecure $\Rightarrow$ a specific leaking observation as witness.
- Two difficulties arise: a *single* probe already asks a question about a distribution over all of $\{0, 1\}^{\mathcal{R}}$,
- and there are combinatorially many probes to settle.

How Hard Is One Probe?

Warm-up: three views of secret-independence

Fix a probe O (a tuple w of wires). Write $\mathcal{L}_a(O)$ for its distribution over a uniform tape $\rho \in \{0, 1\}^{\mathcal{R}}$, and $N_a(o) = |\{\rho : O(\rho; a) = o\}|$ for the tape counts.

1. **Distributional.** $\mathcal{L}_a(O) = \mathcal{L}_b(O)$ for all secrets a, b . *(the definition)*

Warm-up: three views of secret-independence

Fix a probe O (a tuple w of wires). Write $\mathcal{L}_a(O)$ for its distribution over a uniform tape $\rho \in \{0, 1\}^{\mathcal{R}}$, and $N_a(o) = |\{\rho : O(\rho; a) = o\}|$ for the tape counts.

1. **Distributional.** $\mathcal{L}_a(O) = \mathcal{L}_b(O)$ for all secrets a, b . *(the definition)*
2. **Counting.** $N_a(o) = N_b(o)$ for every observation o , i.e., two histograms of integers coincide.
→ *this section*

Warm-up: three views of secret-independence

Fix a probe O (a tuple w of wires). Write $\mathcal{L}_a(O)$ for its distribution over a uniform tape $\rho \in \{0, 1\}^{\mathcal{R}}$, and $N_a(o) = |\{\rho : O(\rho; a) = o\}|$ for the tape counts.

1. **Distributional.** $\mathcal{L}_a(O) = \mathcal{L}_b(O)$ for all secrets a, b . *(the definition)*
2. **Counting.** $N_a(o) = N_b(o)$ for every observation o , i.e., two histograms of integers coincide.
 $\rightarrow$ *this section*
3. **Coupling.** there exists a bijection Φ of the tapes such that $O(\rho; a) = O(\Phi(\rho); b)$.
 $\rightarrow$ *next section*

Theorem (thesis, Ch. 4)

Computing one tape count $N_a(o)$ is $\#P$ -complete (Valiant 1979). And deciding $\mathcal{L}_a(O) = \mathcal{L}_b(O)$ is $C=P$ -complete (Wagner 1986).

Theorem (thesis, Ch. 4)

Computing one tape count $N_a(o)$ is $\#P$ -complete (Valiant 1979). And deciding $\mathcal{L}_a(O) = \mathcal{L}_b(O)$ is $C=P$ -complete (Wagner 1986).

- Hardness: reduction from ExactSat. One secret bit and one probed wire already encodes it. (construction @ the appendix slides)
- So the hardness is intrinsic; no feature of masking causes it.

What the placement tells

- A polynomial time exact decision would collapse the *entire* polynomial hierarchy to P
(due to Toda.)
- Deciding never costs more than counting, i.e., $C=P \subseteq P^{\#P}$.
- A single probe is already in $C=P$, and full verification universally quantifies over exponentially many such instances.

Consequence

We do not compute counts. We certify *constructively*: exhibiting a relabelling of the randomness under which the secrets visibly cancel, giving a witness without counting, at the price of completeness!

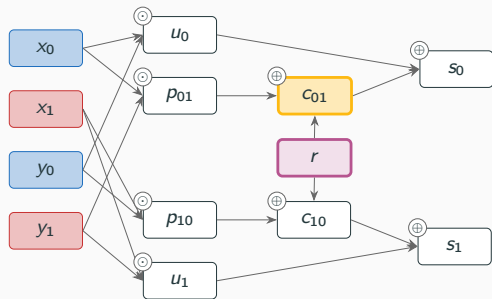
Security Is a Coupling

One substitution, on the running example

Probe the masked cross term: $c_{01} = x_0 \odot y_1 \oplus r$.

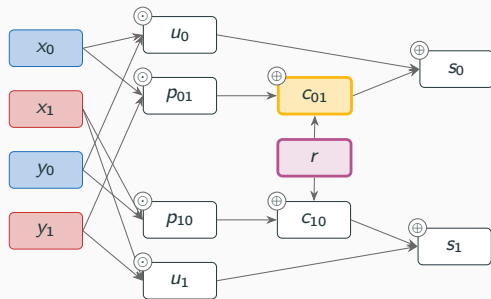
Apply the substitution $\sigma = (r \mapsto x_0 \odot y_1 \oplus r)$:

$$c_{01} \sigma = x_0 \odot y_1 \oplus (x_0 \odot y_1 \oplus r) = r.$$



One substitution, on the running example

Probe the masked cross term: $c_{01} = x_0 \odot y_1 \oplus r$.



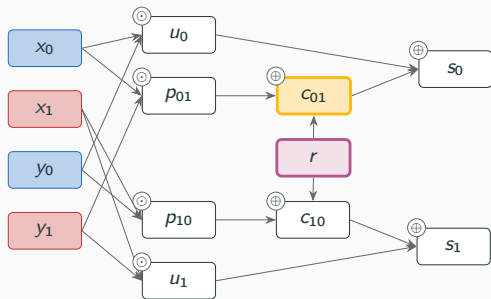
Apply the substitution $\sigma = (r \mapsto x_0 \odot y_1 \oplus r)$:

$$c_{01} \sigma = x_0 \odot y_1 \oplus (x_0 \odot y_1 \oplus r) = r.$$

- The probe is now *syntactically secret-free*: it is uniform, whatever the secrets are.

One substitution, on the running example

Probe the masked cross term: $c_{01} = x_0 \odot y_1 \oplus r$.



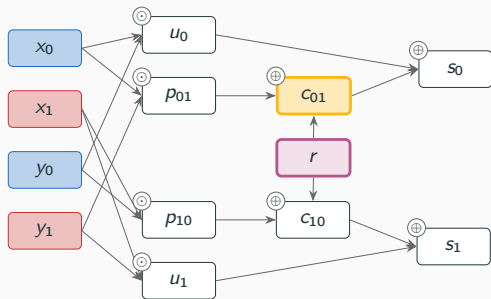
Apply the substitution $\sigma = (r \mapsto x_0 \odot y_1 \oplus r)$:

$$c_{01} \sigma = x_0 \odot y_1 \oplus (x_0 \odot y_1 \oplus r) = r.$$

- The probe is now *syntactically secret-free*: it is uniform, whatever the secrets are.
- And σ induces a *bijection* of the tape space, translating the coordinate r , a *shear*.

One substitution, on the running example

Probe the masked cross term: $c_{01} = x_0 \odot y_1 \oplus r$.



Apply the substitution $\sigma = (r \mapsto x_0 \odot y_1 \oplus r)$:

$$c_{01} \sigma = x_0 \odot y_1 \oplus (x_0 \odot y_1 \oplus r) = r.$$

- The probe is now *syntactically secret-free*: it is uniform, whatever the secrets are.
- And σ induces a *bijection* of the tape space, translating the coordinate r , a *shear*.
- But, the context $x_0 \odot y_1$ mentions a share, so the *one* syntactic step induces a *family* of shears. Fixing the secrets a picks $\hat{\sigma}^a$.

Key identity: $(w\sigma)(\rho; a) = w(\hat{\sigma}^a(\rho); a)$, syntax and semantics agree, for every a .

To prove two distributions equal *without computing them*, link their sources ((Barthe et al. 2017) for probabilistic programs). Our two laws arise from the *same* uniform tape, so we link the tapes.

Definition (coupling function)

A *coupling function* from a to b for a probe O is a bijection $\Phi : \{0, 1\}^{\mathcal{R}} \rightarrow \{0, 1\}^{\mathcal{R}}$ with

$$O(\rho; a) = O(\Phi(\rho); b) \quad \text{for every tape } \rho.$$

- Bijectivity preserves the uniform law; the identity says a run under a is reproduced, observation for observation, under b . Hence we have equal laws.

Theorem

A probe O is secret-independent *iff* for every ordered pair a, b there is a coupling function from a to b for O .

Proof sketch.

- ($\Leftarrow$) A bijection merely re-indexes the tape space, so the counts behind the two laws agree.
- ($\Rightarrow$) Equal laws mean the outcome-fibres $\{\rho : O(\rho; a) = o\}$ come in matching sizes; match them block by block and glue. □

(full proof @ the appendix slides)

Theorem

A probe O is secret-independent *iff* for every ordered pair a, b there is a coupling function from a to b for O .

Proof sketch.

- ($\Leftarrow$) A bijection merely re-indexes the tape space, so the counts behind the two laws agree.
- ($\Rightarrow$) Equal laws mean the outcome-fibres $\{\rho : O(\rho; a) = o\}$ come in matching sizes; match them block by block and glue. □

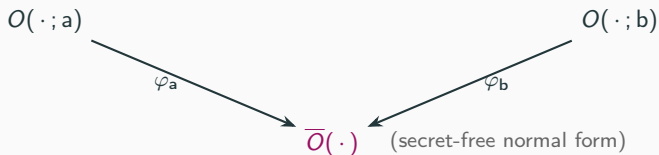
(full proof @ the appendix slides)

This is an existence statement and not a procedure

Matching the fibres needs the counts $N_a(o)$, which are $\#P$ -complete. The theorem guarantees a coupling; it does not find one.

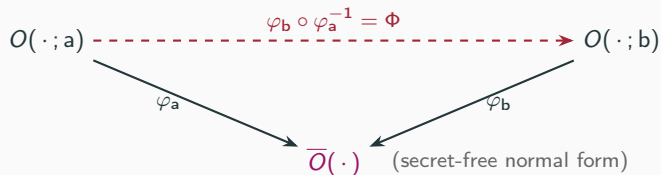
From substitutions to couplings

A derivation $\sigma_1, \dots, \sigma_K$ ending secret-free is a *recipe*: run under secrets a , it becomes the tape bijection $\varphi_a = \widehat{\sigma}_1^a \circ \dots \circ \widehat{\sigma}_K^a$. Applied syntactically, it yields $\overline{O} = O\sigma_1 \dots \sigma_K$.



From substitutions to couplings

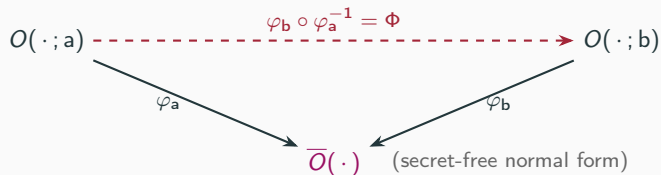
A derivation $\sigma_1, \dots, \sigma_K$ ending secret-free is a *recipe*: run under secrets a , it becomes the tape bijection $\varphi_a = \widehat{\sigma}_1^a \circ \dots \circ \widehat{\sigma}_K^a$. Applied syntactically, it yields $\overline{O} = O\sigma_1 \dots \sigma_K$.



- Every run is a re-labelled view of the *same* secret-free experiment, so any two runs factor through it.

From substitutions to couplings

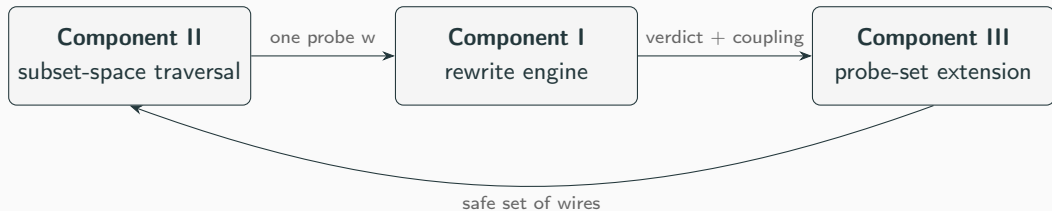
A derivation $\sigma_1, \dots, \sigma_K$ ending secret-free is a *recipe*: run under secrets a , it becomes the tape bijection $\varphi_a = \widehat{\sigma}_1^a \circ \dots \circ \widehat{\sigma}_K^a$. Applied syntactically, it yields $\overline{O} = O\sigma_1 \dots \sigma_K$.



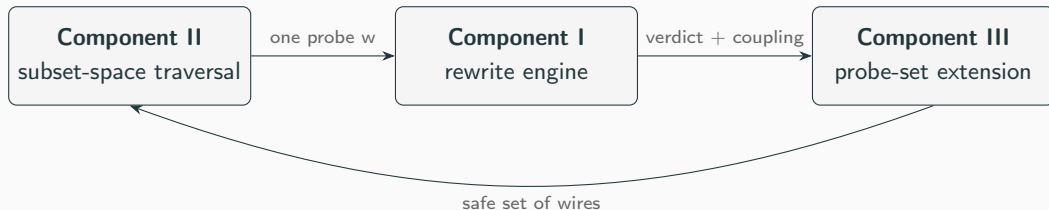
- Every run is a re-labelled view of the *same* secret-free experiment, so any two runs factor through it.
- So a derivation is a **reusable certificate**: the design principle of Coppula.

Coppula: Three Components

Pipeline overview

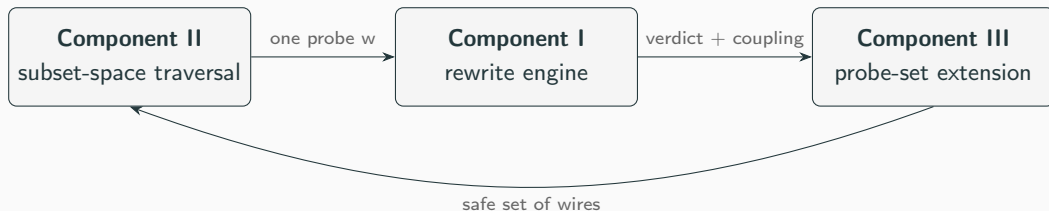


Pipeline overview



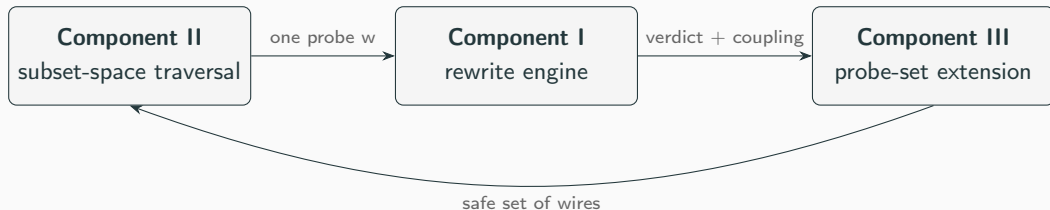
- II sweeps every d -subset without explicitly enumerating them.

Pipeline overview



- I decides a *single* probe, by construction.
- II sweeps every d -subset without explicitly enumerating them.

Pipeline overview



- I decides a *single* probe, by construction.
- II sweeps *every* d -subset without explicitly enumerating them.
- III amplifies one certificate into a whole *safe set*, shrinking the space II must sweep.

Component I: the rewrite engine

Decides one probe w by substituting randoms away, one at a time, driving the observation toward a secret-free form.

- **Tiered.** A cheap single-occurrence rule first (r occurs once across w : substitute, done); with a complete substitution loop behind it.

Component I: the rewrite engine

Decides one probe w by substituting randoms away, one at a time, driving the observation toward a secret-free form.

- **Tiered.** A cheap single-occurrence rule first (r occurs once across w : substitute, done); with a complete substitution loop behind it.
- **Sound.** Every step is a shear, so any Secure verdict carries a genuine coupling (Theorem, Ch. 5).

Component I: the rewrite engine

Decides one probe w by substituting randoms away, one at a time, driving the observation toward a secret-free form.

- **Tiered.** A cheap single-occurrence rule first (r occurs once across w : substitute, done); with a complete substitution loop behind it.
- **Sound.** Every step is a shear, so any Secure verdict carries a genuine coupling (Theorem, Ch. 5).
- **Incomplete, knowingly.** A probe can be independent for reasons no relabelling of this kind captures. This is the price of the hardness result.

Component I: the rewrite engine

Decides one probe w by substituting randoms away, one at a time, driving the observation toward a secret-free form.

- **Tiered.** A cheap single-occurrence rule first (r occurs once across w : substitute, done); with a complete substitution loop behind it.
- **Sound.** Every step is a shear, so any Secure verdict carries a genuine coupling (Theorem, Ch. 5).
- **Incomplete, knowingly.** A probe can be independent for reasons no relabelling of this kind captures. This is the price of the hardness result.

On the example: $c_{01} \xrightarrow{r \mapsto x_0 \odot y_1 \oplus r} r$, one step, Secure, coupling recorded.

Some masks are buried inside *products*, where substitution cannot reach them.

- Substituting $r \mapsto e \oplus r$ needs r to occur as a *summand*; in $u \odot r \oplus u \odot v$ every occurrence of r is trapped inside a product.

Component I: why factoring

Some masks are buried inside *products*, where substitution cannot reach them.

- Substituting $r \mapsto e \oplus r$ needs r to occur as a *summand*; in $u \odot r \oplus u \odot v$ every occurrence of r is trapped inside a product.
- **Multiplicative factoring** pulls the common factor out,

$$u \odot r \oplus u \odot v \longmapsto u \odot (r \oplus v),$$

and the hidden random surfaces as a summand again, ready to substitute.

Component I: why factoring

Some masks are buried inside *products*, where substitution cannot reach them.

- Substituting $r \mapsto e \oplus r$ needs r to occur as a *summand*; in $u \odot r \oplus u \odot v$ every occurrence of r is trapped inside a product.
- **Multiplicative factoring** pulls the common factor out,

$$u \odot r \oplus u \odot v \longmapsto u \odot (r \oplus v),$$

and the hidden random surfaces as a summand again, ready to substitute.

- Applied lazily between substitution attempts, until the probe discharges or no rule applies.

Component II: the representation

Listing all $\sum_{i \leq d} \binom{|\mathcal{W}|}{i}$ probes is hopeless. So we never list them; we describe whole families compactly, as maskVerif does (Barthe et al. 2019).

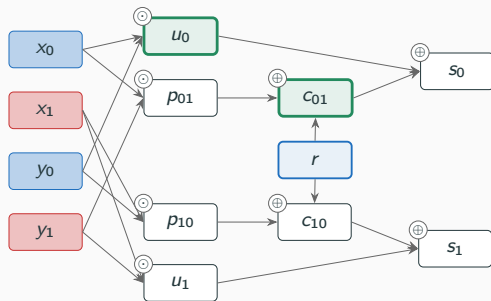
Factors and worklists

- *factor* (c, W) — choose c wires from W .
- *worklist* $\langle (c_1, W_1), \dots, (c_m, W_m) \rangle$, the W_j disjoint — all probes taking c_j from each W_j , $\prod_j \binom{|W_j|}{c_j}$ of them.

DOM-AND, order 2. The eight computed wires $\mathcal{W} = \{u_0, u_1, p_{01}, p_{10}, c_{01}, c_{10}, s_0, s_1\}$, and the whole verification task is the single factor

$$\langle (2, \mathcal{W}) \rangle = \text{all } \binom{8}{2} = 28 \text{ pairs, in one line.}$$

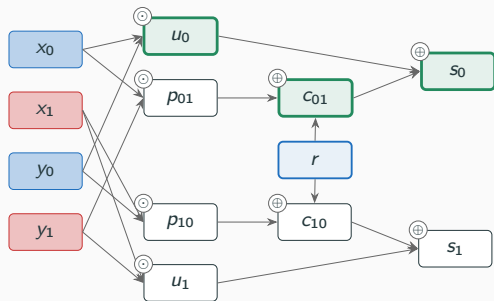
Component II: one recursion step, on our example



the illustrated probe set, in green

1. **Prune?** Discharge the union of all 8 wires at once. It fails here (the gadget leaks at order 2), so we descend.

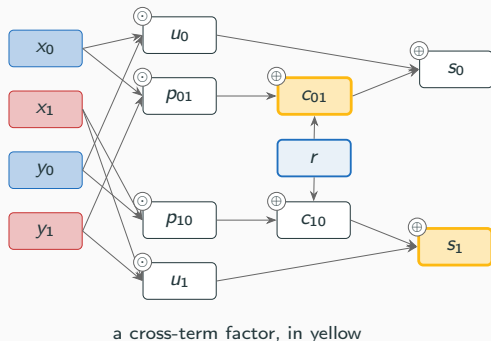
Component II: one recursion step, on our example



the illustrated safe set, in green

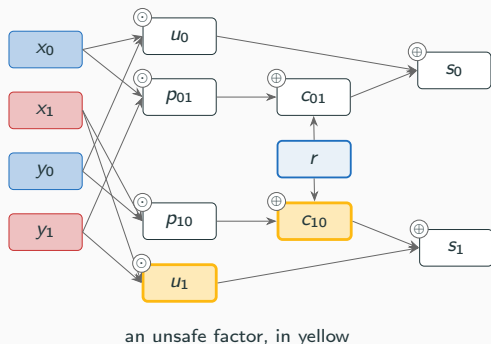
1. **Prune?** Discharge the union of all 8 wires at once. It fails here (the gadget leaks at order 2), so we descend.
2. **Certify a representative probe**, say $\{u_0, c_{01}\}$, and grow it (Component III) into a safe set, here $\{u_0, c_{01}, s_0\}$.

Component II: one recursion step, on our example



1. **Prune?** Discharge the union of all 8 wires at once. It fails here (the gadget leaks at order 2), so we descend.
2. **Certify a representative probe**, say $\{u_0, c_{01}\}$, and grow it (Component III) into a safe set, here $\{u_0, c_{01}, s_0\}$.
3. **Split** along safe / unsafe:
 $28 = \binom{3}{2} + 3 \cdot 5 + \binom{5}{2} = 3 + 15 + 10$ (all-safe, mixed, all-unsafe). *Skip* the all-safe cell and recurse on the 25 that remain.

Component II: one recursion step, on our example

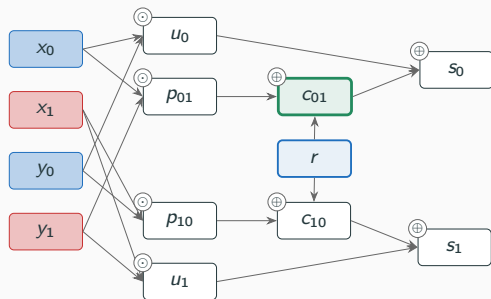


1. **Prune?** Discharge the union of all 8 wires at once. It fails here (the gadget leaks at order 2), so we descend.
2. **Certify a representative probe**, say $\{u_0, c_{01}\}$, and grow it (Component III) into a safe set, here $\{u_0, c_{01}, s_0\}$.
3. **Split** along safe / unsafe:
 $28 = \binom{3}{2} + 3 \cdot 5 + \binom{5}{2} = 3 + 15 + 10$ (all-safe, mixed, all-unsafe). *Skip* the all-safe cell and recurse on the 25 that remain.

Theorem (exhaustiveness, Ch. 5). Read set-wise, Vandermonde's identity says the cells *partition* the probes: every d -subset is visited exactly once.

Component III: growing safe sets

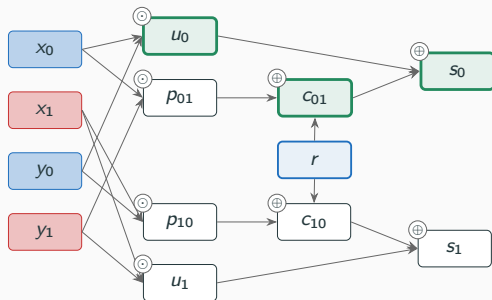
A discharged probe is a coupling, but can the *same* coupling cover more wires?



- A wire u joins the safe set of coupling φ when the composite $u \circ \varphi$ is secret-free.

Component III: growing safe sets

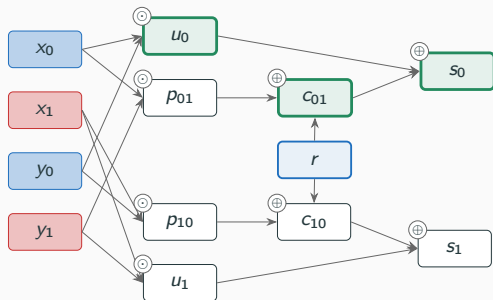
A discharged probe is a coupling, but can the *same* coupling cover more wires?



- A wire u joins the safe set of coupling φ when the composite $u \circ \varphi$ is secret-free.
- Here $\varphi = (r \mapsto r \oplus x_0 \odot y_1)$ also blinds u_0 and s_0 , as $u_0 \circ \varphi = x_0 \odot y_0$ and $s_0 \circ \varphi = x_0 \odot y_0 \oplus r$.

Component III: growing safe sets

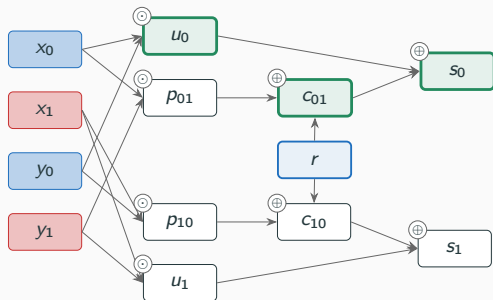
A discharged probe is a coupling, but can the *same* coupling cover more wires?



- A wire u joins the safe set of coupling φ when the composite $u \circ \varphi$ is secret-free.
- Here $\varphi = (r \mapsto r \oplus x_0 \odot y_1)$ also blinds u_0 and s_0 , as $u_0 \circ \varphi = x_0 \odot y_0$ and $s_0 \circ \varphi = x_0 \odot y_0 \oplus r$.
- Exact test via *ANF*: a function depends on a variable iff it occurs in its normal form.

Component III: growing safe sets

A discharged probe is a coupling, but can the *same* coupling cover more wires?



- A wire u joins the safe set of coupling φ when the composite $u \circ \varphi$ is secret-free.
- Here $\varphi = (r \mapsto r \oplus x_0 \odot y_1)$ also blinds u_0 and s_0 , as $u_0 \circ \varphi = x_0 \odot y_0$ and $s_0 \circ \varphi = x_0 \odot y_0 \oplus r$.
- Exact test via *ANF*: a function depends on a variable iff it occurs in its normal form.
- One discharge settles the whole safe set; Component II needs less traversal.

Component III: three mechanisms

- **Closure** — build-time, entropy-style deduction ($H(u \mid w) = 0$). Local deduction on a scattered pool: cheap but weak.

Proposition (nesting, Ch. 5)

$\text{closure} \subseteq \text{replay} \subseteq \text{coupling}$ — every wire the cheaper mechanisms admit, the coupling admits too. Coppula adopts the coupling approach.

Component III: three mechanisms

- **Closure** — build-time, entropy-style deduction ($H(u \mid w) = 0$). Local deduction on a scattered pool: cheap but weak.
- **Replay** — re-run the derivation on the candidate wire (Barthe et al. 2015). An imperfect fit for us because we *factor*, and factoring is not a replayable step.

Proposition (nesting, Ch. 5)

$\text{closure} \subseteq \text{replay} \subseteq \text{coupling}$ — every wire the cheaper mechanisms admit, the coupling admits too. Coppula adopts the coupling approach.

Component III: three mechanisms

- **Closure** — build-time, entropy-style deduction ($H(u \mid w) = 0$). Local deduction on a scattered pool: cheap but weak.
- **Replay** — re-run the derivation on the candidate wire (Barthe et al. 2015). An imperfect fit for us because we *factor*, and factoring is not a replayable step.
- **Coupling** — apply the recorded tape bijection and test the ANF endpoint. Exact, and independent of how the derivation was found.

Proposition (nesting, Ch. 5)

$\text{closure} \subseteq \text{replay} \subseteq \text{coupling}$ — every wire the cheaper mechanisms admit, the coupling admits too. Coppula adopts the coupling approach.

Component III: the test

Given the coupling φ and a candidate wire u from the pool: is $u \circ \varphi$ secret-free?

Phase 1 — fast reject (probabilistic, may only *reject*)

Sample tapes; evaluate $u \circ \varphi$ twice per tape, secrets drawn at random vs. all zero. A disagreement proves $u \circ \varphi$ moves with the secrets: drop u . Agreement proves nothing.

Component III: the test

Given the coupling φ and a candidate wire u from the pool: is $u \circ \varphi$ secret-free?

Phase 1 — fast reject (probabilistic, may only *reject*)

Sample tapes; evaluate $u \circ \varphi$ twice per tape, secrets drawn at random vs. all zero. A disagreement proves $u \circ \varphi$ moves with the secrets: drop u . Agreement proves nothing.

Phases 2–3 — exact ($\mathbb{F}_2$ cancellation, then ANF)

A function depends on a variable iff the variable occurs in its unique multilinear normal form. So admit u exactly when no secret survives in the ANF of $u \circ \varphi$ (Lemma, Ch. 5).

Component III: the test

Given the coupling φ and a candidate wire u from the pool: is $u \circ \varphi$ secret-free?

Phase 1 — fast reject (probabilistic, may only *reject*)

Sample tapes; evaluate $u \circ \varphi$ twice per tape, secrets drawn at random vs. all zero. A disagreement proves $u \circ \varphi$ moves with the secrets: drop u . Agreement proves nothing.

Phases 2–3 — exact ($\mathbb{F}_2$ cancellation, then ANF)

A function depends on a variable iff the variable occurs in its unique multilinear normal form. So admit u exactly when no secret survives in the ANF of $u \circ \varphi$ (Lemma, Ch. 5).

Why blinded wires are safe to add. For admitted u , $u(\varphi_a(\rho); a) = \bar{u}(\rho)$ depends on ρ alone; as φ_a preserves the uniform law, the *joint* law of the safe set is the same under every secret (Theorem, Ch. 5).

Implementation & Evaluation

Coppula is implemented in **Lean 4**.

- **Hash-consed DAG** expression representation with canonicalization, where structural sharing makes equality checks and substitutions cheap.
- **Multiplicative factoring** as a normal-form pass over the DAG, applied lazily.
- **Bit-sliced ANF endpoint test** for the extension: word-parallel evaluation decides secret-freeness of $u \circ \varphi$ exactly.

Coppula is implemented in **Lean 4**.

- **Hash-consed DAG** expression representation with canonicalization, where structural sharing makes equality checks and substitutions cheap.
- **Multiplicative factoring** as a normal-form pass over the DAG, applied lazily.
- **Bit-sliced ANF endpoint test** for the extension: word-parallel evaluation decides secret-freeness of $u \circ \varphi$ exactly.

Design and proofs are implementation-agnostic; these choices simply make algorithms fast.

Forty configurations at orders 1–4: ISW multiplications, DOM and TI ANDs, refresh gadgets, Keccak χ (DOM and TI sharings), and the TI S-boxes and quadratic sharings of Bilgin et al. Every verdict matches the ground truth, and every Insecure comes with a concrete witness observation.

- **DOM-AND** (2 shares): Secure at order 1, the witness pair $\{c_{01}, c_{10}\}$ at order 2 (our running example).
- **Four-share $x \oplus yz$ TI** (Bilgin et al. 2015): certified secure at orders 1 and 2, insecure at order 3. The *expected* behaviour of a first-order threshold implementation.

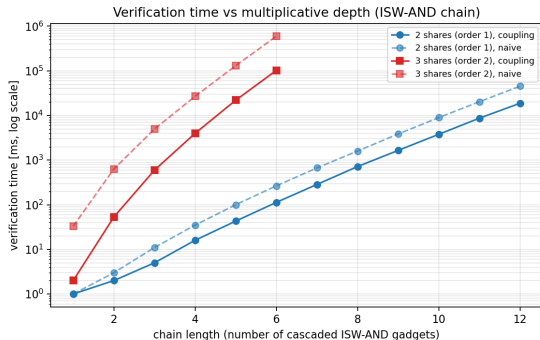
Forty configurations at orders 1–4: ISW multiplications, DOM and TI ANDs, refresh gadgets, Keccak χ (DOM and TI sharings), and the TI S-boxes and quadratic sharings of Bilgin et al. Every verdict matches the ground truth, and every Insecure comes with a concrete witness observation.

- **DOM-AND** (2 shares): Secure at order 1, the witness pair $\{c_{01}, c_{10}\}$ at order 2 (our running example).
- **Four-share $x \oplus yz$ TI** (Bilgin et al. 2015): certified secure at orders 1 and 2, insecure at order 3. The *expected* behaviour of a first-order threshold implementation.
- **The factoring family** ($d+1$ quadratic S-boxes, Q^3/Q^4): randomness-free sharings whose certification must factor first. A factoring-free, maskVerif-style engine false-reports Insecure on five; Coppula verifies every one.

(Full extension-comparison table @ the appendix slides.)

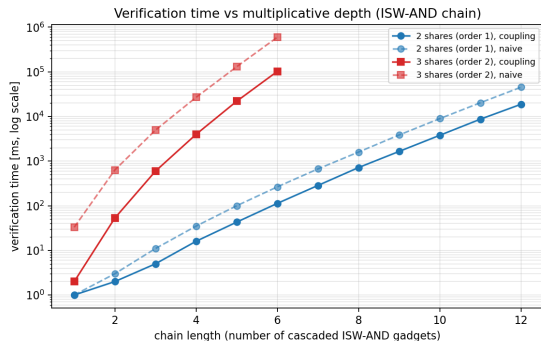
Evaluation: is the extension worth it?

- Coupling matches replay's yield *to the digit* wherever replay is right (five-share ISW: 4735 discharges, 8078 free wires on both), and keeps the correct verdict on the five rows where replay false-reports Insecure.



Verification time versus circuit depth.

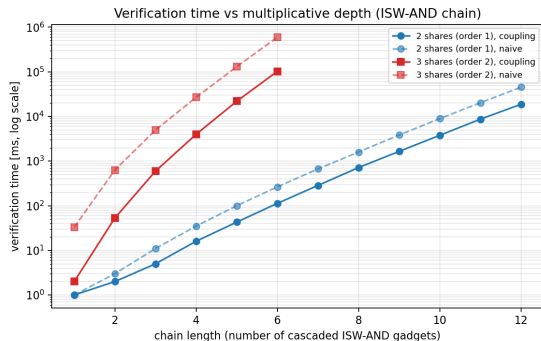
Evaluation: is the extension worth it?



Verification time versus circuit depth.

- Coupling matches replay's yield *to the digit* wherever replay is right (five-share ISW: 4735 discharges, 8078 free wires on both), and keeps the correct verdict on the five rows where replay false-reports Insecure.
- Closures keep the verdicts but lose the yield: 14 vs. 537 free wires on Fides, 632,834 vs. 4735 discharges on five-share ISW, because the split degrades toward enumeration.

Evaluation: is the extension worth it?



Verification time versus circuit depth.

- Coupling matches replay's yield *to the digit* wherever replay is right (five-share ISW: 4735 discharges, 8078 free wires on both), and keeps the correct verdict on the five rows where replay false-reports Insecure.
- Closures keep the verdicts but lose the yield: 14 vs. 537 free wires on Fides, 632,834 vs. 4735 discharges on five-share ISW, because the split degrades toward enumeration.
- The extension pays for itself: Fides certifies 537 wires from 55 discharges; five-share ISW falls from ~ 609 s (no extension) to under 2 s.

Wrap-up

- **Model.** Our model is value-based leakage only: glitches and transitions are out of scope; so is composition. So we show single-circuit, exact probing security.

- **Model.** Our model is value-based leakage only: glitches and transitions are out of scope; so is composition. So we show single-circuit, exact probing security.
- **Larger fields.** The core (shears, couplings, ANF) is already field-agnostic; lifting the implementation beyond $\mathbb{F}_2$ is engineering plus a normal-form choice. (Also, the power of Schwartz–Zippel can come into picture here.)

- **Model.** Our model is value-based leakage only: glitches and transitions are out of scope; so is composition. So we show single-circuit, exact probing security.
- **Larger fields.** The core (shears, couplings, ANF) is already field-agnostic; lifting the implementation beyond $\mathbb{F}_2$ is engineering plus a normal-form choice. (Also, the power of Schwartz–Zippel can come into picture here.)
- **Composing couplings.** for unions of safe sets using couplings as a guide.

- **Model.** Our model is value-based leakage only: glitches and transitions are out of scope; so is composition. So we show single-circuit, exact probing security.
- **Larger fields.** The core (shears, couplings, ANF) is already field-agnostic; lifting the implementation beyond $\mathbb{F}_2$ is engineering plus a normal-form choice. (Also, the power of Schwartz–Zippel can come into picture here.)
- **Composing couplings.** for unions of safe sets using couplings as a guide.
- **More.** smarter factoring heuristics; exploiting circuit symmetries to quotient the subset space.

1. **Placement.** We show that: counting is $\#P$ -complete, deciding is $C=P$ -complete, i.e., they sit above the polynomial hierarchy, ruling out the counting route.

1. **Placement.** We show that: counting is $\#P$ -complete, deciding is $C=P$ -complete, i.e., they sit above the polynomial hierarchy, ruling out the counting route.
2. **A rigorous coupling account of probing security.** Security *is* the existence of a family of coupling functions, the family is a groupoid, and substitutions construct it.

1. **Placement.** We show that: counting is $\#P$ -complete, deciding is $C=P$ -complete, i.e., they sit above the polynomial hierarchy, ruling out the counting route.
2. **A rigorous coupling account of probing security.** Security *is* the existence of a family of coupling functions, the family is a groupoid, and substitutions construct it.
3. **Coppula.** Made up of a sound rewrite engine (with factoring), an exhaustive subset-space traversal, and a probe-set extension with a proven mechanism hierarchy closure $\subseteq$ replay $\subseteq$ coupling.

1. **Placement.** We show that: counting is $\#P$ -complete, deciding is $C=P$ -complete, i.e., they sit above the polynomial hierarchy, ruling out the counting route.
2. **A rigorous coupling account of probing security.** Security *is* the existence of a family of coupling functions, the family is a groupoid, and substitutions construct it.
3. **Coppula.** Made up of a sound rewrite engine (with factoring), an exhaustive subset-space traversal, and a probe-set extension with a proven mechanism hierarchy closure $\subseteq$ replay $\subseteq$ coupling.
4. **Evaluation.** We present exact verdicts, with witnesses, across a suite of masked gadgets, and the coupling extension wins in coverage.

Thank you!

Questions?

Backup

Backup: extension comparison (I/IV)

Each entry: Coupling / Replay / Closure. Verdict S = Secure, I = Insecure; † marks replay's false-Insecures.

Circuit	<i>d</i>	Verdict	Time [ms]	Discharges	Free wires
ISW multiplication					
2 sh.	1	S/S/S	0/0/1	7/7/15	2/2/0
3 sh.	2	S/S/S	3/1/34	34/34/309	31/31/8
4 sh.	3	S/S/S	46/37/3190	338/338/10770	437/437/439
5 sh.	4	S/S/S	1125/845/409472	4735/4735/632834	8078/8078/37513
DOM AND					
2 sh.	1	S/S/S	1/0/0	11/11/15	2/2/0
2 sh.	2	I/I/I	0/0/1	2/2/6	0/0/2
3 sh.	2	S/S/S	5/4/35	88/88/325	44/44/7
4 sh.	3	S/S/S	104/90/3419	859/859/13291	784/784/532

Backup: extension comparison (II/IV)

Circuit	d	Verdict	Time [ms]	Discharges	Free wires
Trichina AND					
well-formed	1	S/S/S	0/1/1	11/11/15	2/2/0
mis-associated	1	I/I/I	1/0/1	10/10/12	1/1/0
TI AND					
3 sh.	1	S/S/S	0/0/1	9/9/27	9/9/0
3 sh.	2	I/I/I	1/1/1	10/10/10	5/5/2
Mask refreshing					
additive, 3 sh.	2	S/S/S	0/0/0	6/6/8	1/1/0
additive, 4 sh.	3	S/S/S	1/0/1	11/11/17	6/6/0
ISW, 3 sh.	2	S/S/S	0/0/0	6/6/10	4/4/0
ISW, 4 sh.	3	S/S/S	0/1/2	12/12/51	24/24/0
Keccak χ					
DOM, 2 sh.	1	S/S/S	14/11/29	49/49/107	29/29/5
DOM, 3 sh.	2	S/S/S	2807/1797/11563	2865/2865/14047	2671/2671/645
TI, 3 sh.	1	S/S/S	21/14/103	43/43/183	68/68/5
TI, 3 sh.	2	I/I/I	21/14/72	57/57/352	92/92/27

Backup: extension comparison (III/IV)

Circuit	d	Verdict	Time [ms]	Discharges	Free wires
TI S-boxes and quadratic sharings					
Q_{12}^4 direct, 2 sh.	1	S/I [†] /S	2/2/5	27/26/47	16/16/10
Q_{12}^4 TI, 3 sh.	1	S/S/S	5/4/25	25/25/83	30/30/6
Fides AB1, 4 sh.	1	S/S/S	1941/972/36461	55/55/1127	537/537/14
$x + yz$ direct, 3 sh.	1	S/S/S	1/0/3	7/7/29	10/10/0
$x + yz$ w/ CT, 3 sh.	1	S/S/S	1/1/3	11/11/29	9/9/0
$x + yz$, 4 sh.	2	S/S/S	11/8/170	78/78/767	115/115/20
$x + yz$, 4 sh.	3	I/I/I	11/7/8	102/102/72	99/99/15
$x + yz + xyz$, 4 sh.	1	S/S/S	45/18/175	31/31/167	63/63/1
xy virtual var.	1	S/S/S	3/1/17	9/9/55	23/23/0
xy virtual share	1	S/S/S	1/0/7	9/9/37	15/15/0
xy , 4 $\rightarrow$ 3 sh.	1	S/S/S	1/1/3	11/11/25	8/8/0

Backup: extension comparison (IV/IV)

Circuit	d	Verdict	Time [ms]	Discharges	Free wires
$d+1$ quadratic S-box family					
Q_1^3	1	S/S/S	0/1/0	13/13/23	11/11/10
Q_2^3	1	S/I ⁺ /S	2/1/4	25/24/43	13/13/6
Q_3^3	1	I/I/I	4/2/7	18/18/32	22/22/2
Q_4^4	1	S/S/S	1/0/1	15/15/27	14/14/14
Q_{12}^4	1	S/I ⁺ /S	2/2/5	25/24/47	15/15/8
Q_{293}^4	1	S/I ⁺ /S	4/2/6	31/24/59	20/20/8
Q_{294}^4	1	S/S/S	1/1/3	23/23/39	16/16/12
Q_{299}^4	1	S/I ⁺ /S	5/2/13	35/18/75	26/22/8
Q_{300}^4	1	I/I/I	4/3/8	18/18/32	24/24/2

The two genuinely insecure classes, Q_3^3 and Q_{300}^4 , are exactly the quadratic classes that admit no uniform three-share TI (Bilgin et al. 2015).

Backup: C=P membership, sketch

Encode disagreement of the two laws as a single gap function:

$$f(C, O, a, b) = \sum_o (N_a(o) - N_b(o))^2.$$

Each N is a #P function; GapP is closed under subtraction and multiplication (Fenner et al. 1994); the sum has $2^k \leq 2^d$ terms, constant for fixed d . A sum of squares vanishes iff every term does, so

$$\mathcal{L}_a(O) = \mathcal{L}_b(O) \iff f = 0,$$

and the problem is the vanishing set of a gap function, which is exactly C=P.

Backup: the ExactSat reduction

Reduce from the equality form of ExactSat: given formulas φ_0, φ_1 over variables $x_1, \dots, x_m$, decide $\#\varphi_0 = \#\varphi_1$ (C=P-complete, (Wagner 1986)).

Build a circuit with one secret bit s , the x_i as randoms, and the single wire

$$w = \varphi_0 \oplus (s \odot (\varphi_0 \oplus \varphi_1)),$$

realising the formulas over $\oplus$ and $\odot$. Then w reads φ_0 when $s = 0$ and φ_1 when $s = 1$, so the one-bit probe $\{w\}$ is secret-independent *iff* $\#\varphi_0 = \#\varphi_1$.

The construction is polynomial, and one secret bit with one probed wire suffices: no feature of masking is what makes the problem hard.

Backup: security is a coupling, full proof

Write $N_a(o) = |\{\rho : O(\rho; a) = o\}|$ for the tape counts.

($\Leftarrow$) Let φ be a coupling function from a to b . For any observation o , the identity $O(\rho; a) = O(\varphi(\rho); b)$ and the bijectivity of φ give, reindexing by $\rho' = \varphi(\rho)$,

$$N_a(o) = |\{\rho : O(\rho; a) = o\}| = |\{\rho : O(\varphi(\rho); b) = o\}| = |\{\rho' : O(\rho'; b) = o\}| = N_b(o).$$

Hence $\mathcal{L}_a(O) = \mathcal{L}_b(O)$; if this holds for every pair, O is secret-independent by definition.

($\Rightarrow$) Suppose $\mathcal{L}_a(O) = \mathcal{L}_b(O)$, i.e. $N_a(o) = N_b(o)$ for every o . The fibres $F_a(o) = \{\rho \mid O(\rho; a) = o\}$ partition $\{0, 1\}^{\mathcal{R}}$ as o ranges over the observations, and likewise the $F_b(o)$. Corresponding blocks have equal size, so pick a bijection $F_a(o) \rightarrow F_b(o)$ for each o and take their union φ . It is a bijection of $\{0, 1\}^{\mathcal{R}}$, and by construction carries each $\rho \in F_a(o)$ to some $\rho' \in F_b(o)$, so $O(\varphi(\rho); b) = o = O(\rho; a)$. Thus φ is a coupling function from a to b . $\square$

Backup: replay vs. factoring

The derivations of Barthe et al. (2015) replay two rule kinds: Opt (substitution) and Conv (ANF normalisation) — for them, ANF is a *replayed rule*.

Our derivations interleave substitutions with *multiplicative factoring*, and a factoring step has no tape-level counterpart to replay. Replaying the substitution steps alone on a new wire can spuriously fail, producing false-Insecure verdicts (observed in the evaluation, the † rows).

Hence the coupling mechanism: apply the recorded bijection $\hat{\sigma}_1 \circ \dots \circ \hat{\sigma}_k$ semantically and test the ANF endpoint — exact, and indifferent to how the derivation was found.

Backup: positioning

- **maskVerif** (Barthe et al. 2019): language-based, the traversal's main inspiration; replay-style extension; no factoring.
- **REBECCA / Fourier-analytic** (Bloem et al. 2018): correlation sets over Fourier coefficients; approximate in general.
- **SILVER** (Knichel et al. 2020): exact, BDD-based model counting; exhaustive per-probe distributions, different scaling profile.
- **Coppula**: exact like SILVER, constructive like maskVerif — and the certificate (the coupling) is itself an object with structure we exploit.

Backup: the four-share TI netlist

Four-share TI of $x \oplus (y \odot z)$ (Bilgin et al. 2015); each input four-shared, sixteen cross products $y_i \odot z_j$:

$$\begin{aligned} F_1 &= x_2 \oplus \bigoplus_{i,j \in \{2,3,4\}} y_i z_j, & F_2 &= x_3 \oplus y_1 z_3 \oplus y_1 z_4 \oplus y_3 z_1 \oplus y_4 z_1 \oplus y_1 z_1, \\ F_3 &= x_4 \oplus y_1 z_2 \oplus y_2 z_1, & F_4 &= x_1. \end{aligned}$$

A *first-order* threshold implementation — never claimed beyond order one. Coppel certifies orders 1 and 2 and finds the order-3 witness: a counterexample on a gadget with a *known* failure order, not a flaw.

-  Barthe, Gilles et al. (2015). **“Verified Proofs of Higher-Order Masking”**. In: *Advances in Cryptology – EUROCRYPT 2015*. Vol. 9056. Lecture Notes in Computer Science. Springer, pp. 457–485. doi: 10.1007/978-3-662-46800-5_18.
-  Barthe, Gilles et al. (2017). **“Coupling Proofs are Probabilistic Product Programs”**. In: *Proceedings of the 44th ACM SIGPLAN Symposium on Principles of Programming Languages (POPL)*. ACM, pp. 161–174. doi: 10.1145/3009837.3009896.
-  Barthe, Gilles et al. (2019). **“maskVerif: Automated Verification of Higher-Order Masking in Presence of Physical Defaults”**. In: *Computer Security – ESORICS 2019*. Ed. by Kazue Sako, Steve A. Schneider, and Peter Y. A. Ryan. Vol. 11735. Lecture Notes in Computer Science. Cham: Springer International Publishing, pp. 300–318. isbn: 978-3-030-29959-0. doi: 10.1007/978-3-030-29959-0_15.
-  Bilgin, Begül et al. (2015). **“Threshold Implementations of Small S-boxes”**. In: *Cryptography and Communications* 7.1, pp. 3–33. doi: 10.1007/s12095-014-0104-7.
-  Bloem, Roderick et al. (2018). **“Formal Verification of Masked Hardware Implementations in the Presence of Glitches”**. In: *Advances in Cryptology – EUROCRYPT 2018*. Vol. 10821. Lecture Notes in Computer Science. Springer, pp. 321–353. doi: 10.1007/978-3-319-78375-8_11.

-  Chari, Suresh et al. (1999). “**Towards Sound Approaches to Counteract Power-Analysis Attacks**”. In: *Advances in Cryptology – CRYPTO '99*. Vol. 1666. Lecture Notes in Computer Science. Springer, pp. 398–412. doi: 10.1007/3-540-48405-1_26.
-  Duc, Alexandre, Stefan Dziembowski, and Sebastian Faust (2014). ***Unifying Leakage Models: from Probing Attacks to Noisy Leakage***. Cryptology ePrint Archive, Paper 2014/079. url: <https://eprint.iacr.org/2014/079>.
-  Fenner, Stephen A., Lance J. Fortnow, and Stuart A. Kurtz (1994). “**Gap-Definable Counting Classes**”. In: *Journal of Computer and System Sciences* 48.1, pp. 116–148. doi: 10.1016/S0022-0000(05)80024-8.
-  Groß, Hannes, Stefan Mangard, and Thomas Korak (2016). “**Domain-Oriented Masking: Compact Masked Hardware Implementations with Arbitrary Protection Order**”. In: *Proceedings of the 2016 ACM Workshop on Theory of Implementation Security (TIS@CCS)*. ACM, p. 3. doi: 10.1145/2996366.2996426.
-  Ishai, Yuval, Amit Sahai, and David Wagner (2003). “**Private Circuits: Securing Hardware against Probing Attacks**”. In: *Advances in Cryptology - CRYPTO 2003*. Ed. by Dan Boneh. Berlin, Heidelberg: Springer Berlin Heidelberg, pp. 463–481. isbn: 978-3-540-45146-4.

-  Knichel, David, Pascal Sasdrich, and Amir Moradi (2020). **“SILVER – Statistical Independence and Leakage Verification”**. In: *Advances in Cryptology – ASIACRYPT 2020*. Vol. 12491. Lecture Notes in Computer Science. Springer, pp. 787–816. doi: 10.1007/978-3-030-64837-4_26.
-  Kocher, Paul, Joshua Jaffe, and Benjamin Jun (1999). **“Differential Power Analysis”**. In: *Advances in Cryptology – CRYPTO '99*. Vol. 1666. Lecture Notes in Computer Science. Springer, pp. 388–397. doi: 10.1007/3-540-48405-1_25.
-  Valiant, Leslie G. (1979). **“The Complexity of Computing the Permanent”**. In: *Theoretical Computer Science* 8.2, pp. 189–201. doi: 10.1016/0304-3975(79)90044-6.
-  Wagner, Klaus W. (1986). **“The Complexity of Combinatorial Problems with Succinct Input Representation”**. In: *Acta Informatica* 23.3, pp. 325–356. doi: 10.1007/BF00289117.